TECHNICAL DEVELOPMENT REPORT / 01

SILICAFLUX

Production-grade silicon from high-purity silica



Purpose
Define a technically credible route to manufacture electronic-grade polysilicon from high-purity quartz, with clear production gates, impurity controls, energy and material balances, and a staged investment pathway.

Prepared for	Technical scope	Date
SilicaFlux development programme	High-purity silica to electronic-grade silicon	2 November 2026

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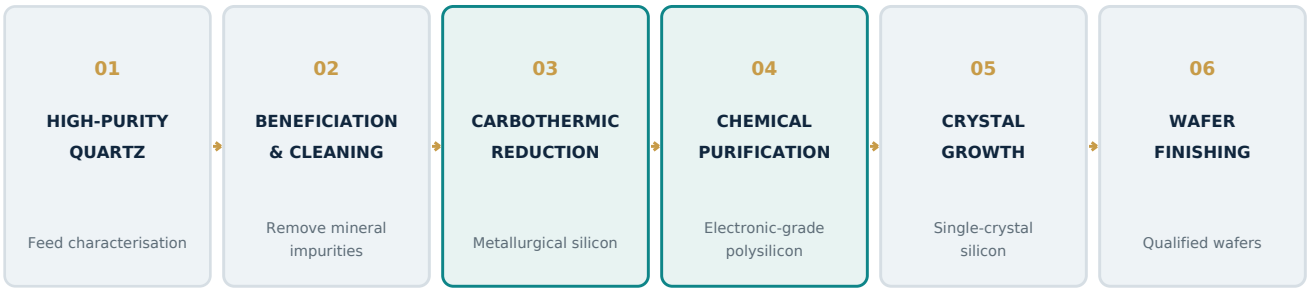
This is a concept-level technical report, not a construction package, process safety design, guaranteed product specification or investment-grade feasibility study. Final design requires representative ore testing, pilot campaigns, customer specifications, vendor engineering, HAZOP/LOPA and environmental permitting.

Executive summary

SilicaFlux should target electronic-grade polysilicon (EG-Si) as its primary semiconductor-material product. The most credible first-generation production route is to beneficiate a carefully characterised quartz feed, reduce it to metallurgical-grade silicon in a submerged-arc furnace, convert that silicon into volatile chlorosilanes, purify those compounds by distillation, and deposit high-purity silicon using a modified Siemens chemical-vapour-deposition (CVD) process.

The critical insight is that exceptional silica feed quality is valuable, but it does not by itself create semiconductor-grade silicon. The reduction furnace and its carbonaceous reductants can add impurities; boron and phosphorus are particularly important because their behaviour through refining is not identical to that of many metallic contaminants. Electronic-grade purity is achieved by controlling the chemistry and purification train, then proving performance through analytical and crystal-growth qualification.

Recommended production route



Core production route: established industrial unit operations; each stage has a separate release specification.

Recommended strategy

Decision	Recommendation	Reason
Feedstock	Selectively mined, fully characterised high-purity quartz; benchmark against lower-cost quartz.	The premium is justified only if it improves recovery, purification duty or final yield.
Reduction	Established submerged-arc carbothermic reduction for initial production.	Commercially mature and scalable; creates metallurgical-grade silicon, not final EG-Si.
Purification	Modified Siemens / trichlorosilane distillation and CVD as the reference route.	Most defensible route for high-purity polysilicon with industrial precedent.
Innovation	Parallel pilot R&D; on molten-salt electrochemical reduction.	Potentially lower-carbon route, but purity, scale and economics remain unproven at commercial EG-Si scale.
Commercialisation	Outsource ingot/wafer qualification and chip fabrication initially.	Avoid coupling a new feedstock process with a new wafer fab at the same time.

Investment principle: prove the quartz-to-silicon impurity balance and qualified polysilicon output before committing to an integrated full-scale plant. The highest-value first asset is a disciplined pilot and analytical capability, not a large furnace.

Report structure and product definition

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2. Product definition

In this report, “production-grade silicon” means electronic-grade polysilicon suitable as a feedstock for semiconductor single-crystal growth. It does not mean that the output is already a semiconductor device or a finished wafer. The material must still pass a customer-specific chemical specification and crystal-growth qualification.

Material grade	Typical role	SilicaFlux position
Metallurgical-grade silicon (MG-Si)	Alloys, silicones and intermediate feed for further refining; commonly around 98–99% Si.	Intermediate product from the reduction furnace. Not suitable as chip-grade feedstock.
Solar-grade silicon	Photovoltaic applications; purity and impurity tolerances differ from semiconductor requirements.	Potential secondary market, subject to specific buyer qualification.
Electronic-grade polysilicon (EG-Si)	Feedstock for semiconductor crystal growth; often described using 9N–11N shorthand, but elemental limits matter more.	Primary target product. Must meet an agreed customer specification, not merely a nominal N-number.
Monocrystalline silicon	Single-crystal ingot for wafer manufacture.	Later-stage product after CZ or specialist FZ crystal growth and qualification.

A “9N” or “11N” purity label is only a summary. It can conceal an unacceptable concentration of one electrically active impurity. SilicaFlux should sell against a full elemental impurity schedule, sampling protocol, analytical method and lot-release certificate agreed with the crystal grower or wafer customer.

3. High-purity quartz: selection and qualification

The feedstock decision should begin with geology and impurity deportment, not a headline SiO_2 percentage. Quartz deposits can contain impurities as surface coatings, separate mineral inclusions, fluid inclusions or lattice-bound substitutions. These forms respond differently to crushing, physical separation, thermal treatment and chemical leaching. Aluminium, iron, titanium, alkalis, calcium, boron and phosphorus are among the elements that merit attention. [1][2][3]

High-purity quartz is commonly described in the literature and USGS material as at least 99.99% SiO_2 (less than about 100 ppm total impurities); ultra-high-purity material may contain 10 ppm or less total impurities. These labels are not universal semiconductor feed specifications. [1]

Indicative geological screening benchmarks

A published review cites a commonly used research benchmark for individual quartz grains: total selected impurities below about 50 ppm, with indicative element limits shown below. These are screening references, not a guaranteed product standard or a substitute for testing the actual ore body. [2]

Element	Indicative screening level	Why it is monitored
Aluminium (Al)	<30 ppm	Common lattice-related impurity; may be difficult to remove if structurally incorporated.
Titanium (Ti)	<10 ppm	May occur in inclusions or lattice-related forms.
Iron (Fe)	<3 ppm	Metallic contamination; can affect silicon electrical and crystal quality.
Calcium (Ca)	<5 ppm	May be associated with feldspar/carbonate inclusions.
Lithium (Li)	<5 ppm	May occur as a charge-balancing ion in quartz.
Sodium / potassium	<8 ppm each	Alkali contamination; monitor both mineral and process sources.
Phosphorus (P)	<2 ppm	Potential electrical dopant; requires dedicated analysis.
Boron (B)	<1 ppm	Potential p-type dopant; particularly important in silicon refinement.

The exact assay must distinguish bulk chemistry from single-grain chemistry and, where possible, distinguish surface/inclusion contamination from lattice-bound impurities. Do not reject or accept a deposit on one bulk XRF result alone.

Sampling programme

- Collect representative samples across geological domains, benches and depth intervals; retain duplicate and referee samples.
- Use petrography, automated mineralogy and SEM-EDS to identify inclusions and mineral associations.
- Use ICP-MS/ICP-OES after validated digestion or fusion for trace-element chemistry; use dedicated methods for boron and phosphorus.
- Test multiple size fractions because liberation and impurity deportment can vary with particle size.
- Run pilot beneficiation and downstream silicon reduction on representative composites before reserving a mine block as “semiconductor feed”.

4. Beneficiation and cleaning of silica

The purpose of beneficiation is to remove contamination before reduction, where practical and economically sensible. The route should be selected from mineralogical testwork. More processing is not automatically better: aggressive crushing can create new contamination, and chemical treatments can add their own residues or generate difficult effluents. [3][4]

Unit operation	Purpose	Design notes / limitations
Selective mining and hand/optical sorting	Exclude visibly altered, stained or mineralised quartz zones.	Often the cheapest impurity removal; requires grade control and traceability.
Low-contamination crushing and screening	Liberate inclusions and define a controlled size distribution.	Choose wear materials carefully; prevent iron/steel pickup and excess fines.
Scrubbing / attrition	Remove surface coatings and weakly attached particles.	Water quality and equipment wear become potential contamination sources.
Magnetic separation	Remove iron-bearing and other magnetic/paramagnetic minerals.	Effectiveness depends on impurity mineralogy and liberation size.
Flotation / gravity separation	Separate feldspar, mica and other associated gangue minerals.	Reagent residues and process-water loops require control.
Thermal treatment / quench	May open inclusions or improve subsequent leaching.	Energy-intensive; test whether it produces measurable impurity removal.
Chemical leaching	Remove accessible surface and mineral-bound contaminants.	Acid selection, corrosion, effluent treatment and worker exposure are major design constraints. HF-based processes require specialist controls.
Rinse, dry, classify and pack	Remove residues and prevent recontamination.	Use validated water quality, clean materials handling and controlled packaging.

Recommended process-development approach

Start with a simple flowsheet (sorting, controlled crushing, scrubbing, magnetic separation and classification). Add flotation, thermal treatment or chemical leaching only where mineralogy and mass-balance data show a clear incremental benefit. Each added stage should be evaluated by impurity reduction, silica recovery, water/reagent use, energy, waste generation and cost per tonne of accepted concentrate.

Critical control: avoid introducing the impurities you are trying to remove. Use non-contaminating or qualified wear surfaces where justified, high-purity process water for final rinsing, clean transfer equipment and batch segregation. A final concentrate assay must be tied to the same batch ID used in later silicon production.

Deliverable from this stage

A defined HPQ concentrate specification, including SiO_2, total and element-specific impurities, particle-size distribution, moisture, inclusion mineralogy, bulk density, packaging and sampling method. It should also include an agreed rejection/hold/reprocess protocol.

5. Carbothermic reduction to metallurgical silicon

The industrial reference route is submerged-arc-furnace carbothermic reduction. Silica is reacted with a controlled carbonaceous burden at high temperature to produce metallurgical-grade silicon (MG-Si), with carbon monoxide-rich furnace gas. The simplified overall net reaction is:

$$SiO_2 + 2\,C \rightarrow Si + 2\,CO$$

The actual furnace chemistry is more complex than this single equation. It involves intermediate silicon monoxide and silicon carbide reactions, gas-solid transport, slag formation, electrode behaviour and a vertically varying thermal/reaction profile. The reduction route is mature and commercially used; reviews report typical specific electricity consumption around 11–13 kWh per kg of silicon, although actual performance depends on furnace, burden, electricity and boundary definition. [5][6]

Feed and burden design

- Qualified quartz concentrate with a controlled size distribution and low fines content.
- Carbon reductants selected by fixed carbon, ash, volatile matter, electrical resistivity, reactivity, size strength and impurity profile.
- Electrode and furnace consumables included in the contamination and operating-cost balance.
- Burden permeability and gas flow controlled to maintain stable furnace operation and avoid excessive silicon losses to off-gas/dust.
- Slag, tapping, metal handling and casting designed to limit contamination and maximise recoverable silicon.

Theoretical material balance for pure silica

Item	Stoichiometric value per 1,000 kg pure SiO_2	Interpretation
Pure SiO_2 feed	1,000 kg	Idealised silica basis; real concentrate contains impurities and moisture.
Carbon consumed	about 400 kg	Chemical minimum from the net reaction; industrial burden requirement differs.
Elemental Si produced	about 467 kg	Theoretical maximum before process losses and impurities.
CO generated	about 933 kg	Theoretical net reaction product; real off-gas contains other species and dust.

These figures are stoichiometric calculations, not a furnace operating recipe or guaranteed yield. Actual reductant consumption is influenced by furnace chemistry and losses. The operating model must quantify silicon recovery, slag and dust losses, off-gas composition, electrode consumption, carbon balance and energy use through a pilot furnace campaign.

Furnace operating controls and data

Track electrical energy per tonne of tapped silicon; quartz and reductant feed rates; burden chemistry; electrode consumption; furnace resistance/power stability; tapping temperature; slag mass and composition; off-gas CO/CO_2; dust capture; silicon recovery; and final metal impurity profile. Off-gas collection and treatment are essential, both for safety and for energy/environmental performance.

Output from reduction

The furnace output is MG-Si, commonly around 98–99% silicon. It is an intermediate chemical feed, not electronic-grade silicon. The quality target for MG-Si should be set by the downstream chlorosilane conversion and distillation system, with particular attention to boron, phosphorus, iron, aluminium, calcium and other trace contaminants.

6. Chemical purification to electronic-grade polysilicon

The recommended reference process is a chlorosilane route, normally based on trichlorosilane (TCS) and modified Siemens CVD. It is energy- and capital-intensive, but it is the most defensible first-generation option for a project seeking semiconductor-grade polysilicon. A 2025 review discusses continuing optimisation of the modified Siemens and silane methods, including precursor synthesis, distillation and deposition. [7]

Process sequence

Step	Process description	Control objective
1. MG-Si conditioning	Prepare the silicon feed to the precursor synthesis plant; manage size, surface condition and feed impurities.	Stable conversion and low contaminant carry-through.
2. TCS synthesis	React silicon with hydrogen chloride to form trichlorosilane and hydrogen-containing process streams.	Conversion, selectivity, impurity partitioning and safe containment.
3. Fractional distillation	Separate TCS from higher/lower-boiling chlorosilanes and trace impurities using a multi-column purification train.	Element-specific impurity removal; particularly challenging dopant contaminants.
4. CVD deposition	Decompose purified TCS with hydrogen in a reactor and deposit silicon onto heated silicon starter rods.	Deposit purity, deposition rate, rod quality, contamination control and energy efficiency.
5. Product finishing	Break, clean, classify and package polysilicon in controlled clean conditions.	Prevent downstream contamination and preserve lot traceability.
6. Recycle and recovery	Recover and recycle suitable hydrogen and chlorosilane streams; treat purge and waste streams.	Reduce raw material consumption, emissions and operating cost.

Core reactions (simplified)

Trichlorosilane formation is commonly represented as:
$Si + 3\,HCl \rightarrow HSiCl_3 + H_2$

A simplified silicon deposition reaction is:
$2\,HSiCl_3 + 2\,H_2 \rightarrow 2\,Si + 6\,HCl$

The real plant contains multiple chlorosilane species, recycle loops and equilibrium constraints. The final process design must be based on a vendor-validated process package, reaction engineering, materials compatibility and detailed mass/energy balances.

Why distillation and impurity speciation matter

An overall “ppm of impurities” value is insufficient. The process must understand how each contaminant partitions between the solid silicon, volatile chlorosilanes, distillation cuts, reactor deposits and recycled streams. Boron- and phosphorus-containing species require dedicated attention because they can influence electrical properties at very low concentrations. Set specifications at each process boundary, not just at final packaging.

Alternative: silane / fluidised-bed reactor

A silane-based fluidised-bed reactor (FBR) route can offer continuous granular deposition and potential energy or cost advantages in some product segments. It should be evaluated against the intended semiconductor customer specification and demonstrated product quality. Do not assume an FBR product is interchangeable with rod-deposited Siemens polysilicon without direct qualification.

7. Product qualification and quality assurance

SilicaFlux needs a stage-gated quality system, with independent hold/release points. Each lot should carry a unique identifier linking the mine block, beneficiation campaign, furnace batch, chemical purification campaign, polysilicon lot and subsequent crystal-growth result.

Stage	Recommended analytical methods	Release decision
Quartz ore / concentrate	XRD, petrography, SEM-EDS, ICP-MS/ICP-OES, particle size, moisture; dedicated B/P analysis.	Accept, blend, reprocess or reject feed block / concentrate.
MG-Si	GDMS and/or ICP-MS after validated preparation; selected metals, B, P, Ca, Al, Fe, Ti; carbon/oxygen as relevant.	Release to chlorosilane synthesis only if downstream unit can manage impurity load.
Purified chlorosilanes	Validated trace-element and chemical-speciation analyses; moisture and particle/metal contamination checks where relevant.	Release by column / batch against process-specific limits.
Electronic-grade polysilicon	GDMS for trace elements; dedicated B/P methods; surface/particle/packaging contamination checks.	Release only against customer-approved elemental limits and sampling plan.
Monocrystalline ingot	Resistivity and dopant mapping; FTIR for oxygen/carbon; defect and dislocation inspection; axial/radial uniformity.	Release to wafering only if crystal-growth performance is acceptable.
Finished wafer	Thickness, flatness, TTV, surface particles, roughness, resistivity, crystal defects and contamination.	Customer qualification / wafer specification.

Key technical distinction

Chemical purity, electrical quality and crystal quality are related but not identical. A high-purity polysilicon sample can still produce poor crystals if oxygen, carbon, dopant uniformity, inclusions or growth conditions are not controlled. A polysilicon release certificate is necessary but not sufficient: crystal growth is the decisive downstream test.

Recommended quality architecture

- Use independent third-party laboratory cross-checks during pilot operations.
- Maintain blank, duplicate, reference and spike samples for critical analytical methods.
- Define detection limits below the customer acceptance limits; “not detected” is meaningful only when the method detection limit is stated.
- Use statistical process control for recurring lots and retain samples for root-cause analysis.
- Create a batch genealogy system linking geological source to crystal qualification.

Initial acceptance specification should be negotiated with a target semiconductor crystal grower. Do not publish a generic 11N guarantee before an accredited analytical programme and repeatable full-scale lot history support it.

8. Crystal growth and wafer validation

Once polysilicon meets its chemical specification, it is melted and grown into single-crystal silicon. For a first commercial product, the Czochralski (CZ) method is the logical volume route. A seed crystal is brought into contact with molten silicon and slowly withdrawn while rotation and thermal conditions are controlled to grow a cylindrical single crystal.

High-purity fused-quartz crucibles are used in CZ growth. The crucible is not perfectly inert: oxygen can enter the melt and become incorporated into the crystal. This makes the quality of the quartz crucible, coatings, melt environment and growth conditions a critical part of silicon quality. USGS notes that ultra-high-purity quartz is used in fused-quartz crucibles for semiconductor manufacturing. [1]

CZ process control priorities

- Starting charge quality and loading cleanliness.
- Crucible purity, geometry, surface condition and supplier qualification.
- Temperature field, heater power, pull rate, rotation and melt level.
- Dopant addition and resistivity uniformity, where doped crystal is required.
- Oxygen and carbon incorporation; dislocation and vacancy/interstitial defect control.
- Ingot diameter, axial/radial uniformity, yield and reproducibility.

Float-zone (FZ) growth

Float-zone growth avoids a silica crucible and can deliver very low oxygen concentrations for specialist applications. It is more technically demanding and often less suited to high-throughput, large-diameter production. Treat FZ as a premium/specialty R&D; or partner route rather than the baseline commercial flow.

Wafer finishing

The ingot is cropped, ground, oriented and sliced into wafers, followed by edge shaping, lapping or grinding, polishing, cleaning and inspection. Kerf loss from slicing is a material-efficiency issue. SilicaFlux should investigate recovery and recycling of silicon kerf and offcuts, subject to contamination and economic assessment. Recovered silicon can be a valuable secondary feed, but must be segregated and requalified before re-entry.

Do not call the product a qualified semiconductor wafer until customers have approved the wafer specification, surface quality, electrical characteristics and process performance. The first SilicaFlux product should be sold as electronic-grade polysilicon unless wafer qualification has been achieved.

9. Material, energy and carbon model

A preliminary process model should be constructed around a fixed sales target for accepted electronic-grade polysilicon. Work backwards through each unit operation using measured recoveries rather than assuming that the theoretical silica-to-silicon conversion equals plant output.

Theoretical chemistry baseline

For the net carbothermic reaction, 1,000 kg of pure SiO_2 contains about 467 kg of silicon. The stoichiometric carbon requirement is about 400 kg, and the net equation produces about 933 kg of CO. This is a theoretical benchmark only; it excludes excess reductant, intermediate reactions, slag/dust losses, electrode consumption, moisture, impurities and recycle.

Industrial energy reference points

Process section	Indicative published reference	Use in SilicaFlux model
MG-Si submerged-arc reduction	Approximately 11–13 kWh/kg Si electricity in published reviews. [5][6]	Benchmark for screening; replace with vendor guarantee and pilot data.
Modified Siemens polysilicon	Can be highly energy intensive; published estimates vary substantially with process boundary and plant configuration. [5][7]	Request full utility balance from technology licensor; separate electricity, steam, cooling, hydrogen and recovery loads.
Crystal growth and wafering	Highly dependent on diameter, yield, puller design, wafer thickness and kerf.	Model per kg of accepted ingot and per saleable wafer, not per kg charged.

The major economic issue

The expensive part of the chain is not simply reducing silica to silicon. The chemical purification and crystal-growth stages can dominate energy use, capital intensity and product qualification effort. This is why the feedstock premium for high-purity quartz must be proven by downstream benefit: higher silicon recovery, lower distillation load, fewer off-spec lots, better crystal yield or a lower total cost per accepted kilogram of EG-Si.

Proposed spreadsheet model inputs

Variable	Definition	Evidence required
Q_feed	Tonnes of dry silica concentrate per period.	Mine plan, beneficiation yield and moisture data.
R_benef	Silica recovery through beneficiation.	Pilot mass balance by size fraction and discard stream.
R_furnace	Recoverable silicon from furnace silicon-bearing feed.	Pilot furnace campaign, slag and dust assays.
R_purif	MG-Si converted to accepted EG-Si.	Chlorosilane conversion, distillation and CVD mass balance.
R_crystal	EG-Si converted into accepted crystal/wafer material.	Crystal growth and wafering yield data.
E_total	Total energy use across the chosen boundary.	Metered electricity, fuel, steam, cooling and gas consumption.
C_intensity	CO_2e per accepted kilogram of product.	Direct emissions, electricity factor, reductants, logistics and chemicals.

A proper techno-economic model must state the system boundary: mine-to-MG-Si, mine-to-EG-Si, or mine-to-wafer. Do not compare energy or carbon figures from different boundaries as if they were like-for-like.

10. Alternative routes and SilicaFlux innovation

The proven route should carry the first production programme. Innovation should be managed as a parallel development portfolio with explicit readiness gates, not embedded prematurely into the first plant design.

Route	Potential benefit	Main challenge	SilicaFlux decision
Carbothermic + Siemens	Established industrial route; clear reference for EG-Si.	Energy intensity, carbon reductant, complex chlorosilane process.	Primary production route.
Silane / FBR	Continuous granular deposition; potential productivity and energy advantages.	Product morphology and purity must meet target semiconductor customer acceptance.	Partner study / second-generation option.
Molten-salt electrochemical reduction	Potential route to reduce carbon use and simplify reduction chemistry.	High-purity product, cell materials, salt management, scale and economics remain unresolved.	Research pilot only; do not base initial capacity on it.
Metallothermic reduction	Potential lower-temperature or alternative reductant pathway.	Reductant cost, by-product separation and contamination; scale-up.	Laboratory screening only.
Upgraded metallurgical routes	May deliver solar-grade or selected specialty silicon with lower energy.	Not automatically suitable for semiconductor EG-Si requirements.	Potential secondary product line, not core chip-grade feed.

Molten-salt electrolysis: R&D; hypothesis

Electrochemical reduction uses electrons as a reducing agent and may avoid the carbon requirement of the conventional furnace route. Research reviews describe silica electro-deoxidation and silicon electrowinning in molten salts, but identify challenges in producing highly pure silicon and developing carbon-free oxygen-evolving anodes. A 2024 review characterises electrochemical and metallothermic options as promising but, at that time, demonstrated mainly at laboratory scale. [5][8]

SilicaFlux experimental questions

- Can an electrochemical cell accept beneficiated natural quartz directly, or does the silica need conversion to a controlled form?
- What cell and electrolyte contamination enters the silicon product, and how is it removed?
- Can the process separate boron and phosphorus adequately for electronic-grade targets?
- Can oxygen be evolved safely and efficiently without a consumable carbon anode?
- What are current efficiency, cell life, salt recycle, energy use and cost per kilogram of accepted silicon?

Innovation rule

Only promote the electrochemical route from R&D; into a commercial plant after repeatable campaigns demonstrate product purity, mass balance, cell lifetime, energy use, environmental control and a competitive cost per kilogram of accepted product.

11. Plant configuration, utilities and safety

A fully integrated plant is a complex chemical and metallurgical site. SilicaFlux should design it as a set of independently containable process areas with material traceability and defined transfer specifications between them.

Proposed process blocks

Area	Principal plant	Key interfaces
A. Mine / quartz preparation	Selective mining, sorting, crushing, screening and stockpiles.	Ore grade control; dust collection; clean feed logistics.
B. Beneficiation	Scrubbers, magnetic separators, flotation where required, leach/rinse, water treatment and drying.	Process water, reagent storage, residues and concentrate packaging.
C. Reduction	Submerged-arc furnace, raw-material handling, tapping/casting, gas capture and off-gas treatment.	High-power electrical supply, reductants, CO-rich off-gas, slag and dust.
D. Chemical purification	MG-Si preparation, chlorosilane synthesis, distillation, CVD reactors, gas recovery and scrubbing.	Hydrogen, HCl, chlorosilanes, cooling, high-purity utilities and hazardous-area controls.
E. Product finishing	Polysilicon breaking, cleaning, grading, clean packaging and sample preparation.	Cleanroom-like handling; traceability; independent QC.
F. Optional crystal / wafer line	CZ pullers, ingot finishing, wire saws, grinding and polishing.	High-purity quartz crucibles, clean utilities, wafer inspection.

Utilities to plan early

- Firm high-voltage electrical connection, power-quality management and furnace electrical infrastructure.
- Cooling water and heat rejection; high-purity process water for relevant downstream operations.
- Hydrogen, HCl and chlorosilane storage, distribution, recovery and abatement systems sized by licensor engineering.
- Nitrogen and other inert gases, compressed air, vacuum, steam and emergency power as required.
- Wastewater treatment, hazardous waste storage, dust collection, off-gas scrubbing and continuous emissions monitoring.

Critical hazards and design controls

Hazard	Why it matters	Minimum design response
Respirable crystalline silica	Dust exposure during mining, crushing, drying and transfer.	Enclosure, local exhaust ventilation, wet methods where appropriate, exposure monitoring and respiratory protection programme.
Carbon monoxide	Toxic furnace off-gas; odour is not a reliable warning.	Gas-tight capture, fixed and personal detection, ventilation, interlocks and emergency response.
Hydrogen / chlorosilanes / HCl	Flammable, corrosive or reactive chemical process streams.	Closed systems, compatible materials, gas detection, segregation, scrubbers/abatement and engineered emergency isolation.
HF or aggressive leach chemistry	Severe acute toxicity/corrosion and difficult effluent management.	Avoid unless justified; specialist plant, trained operators, engineered containment and emergency medical protocols.
High temperature / electrical energy	Furnace, molten silicon, hot surfaces and arc-flash hazards.	Remote operation, guarding, interlocks, safe tapping systems, refractory engineering and electrical safety systems.

12. Development roadmap and scale-up gates

The development programme should reduce uncertainty in sequence. The first question is whether the selected quartz produces better accepted silicon at an attractive total cost—not whether a large integrated plant can be drawn on paper.

Phase	Work programme	Gate / deliverable	Release criterion
0. Feedstock diligence	Geological sampling, mineralogy, chemistry, representative composites, benchmark competing quartz sources.	Qualified feedstock dossier and preliminary impurity deportment map.	Sampling is representative and assay methods are validated.
1. Beneficiation pilot	Pilot separation flowsheet; compare physical-only and selected chemical/thermal options.	Repeatable concentrate quality, recovery, water/reagent and residue balances.	Concentrate meets agreed reduction feed specification at acceptable cost.
2. Reduction pilot	Partner or pilot submerged-arc campaigns using candidate quartz and reductants.	MG-Si samples, impurity mass balance, silicon recovery, energy and gas data.	B/P and metallic impurities are compatible with downstream purification economics.
3. Purification demonstration	Use a specialist chlorosilane/Siemens partner or demonstration facility.	Repeated EG-Si lots with independent analytical verification.	Customer-grade element limits and reliable batch reproducibility are achieved.
4. Crystal qualification	Grow test crystals; characterise resistivity, oxygen, carbon, defects and yield.	Crystal-grower acceptance and documented qualification lot.	Material consistently produces acceptable crystal properties.
5. FEED / commercial design	Licensor selection, front-end engineering, utilities, HAZOP, permitting, offtake.	Investment-grade project package and binding performance guarantees.	Economics remain positive under downside yield, energy and price cases.
6. Commercial ramp	Commissioning, qualification lots, customer audits, ramp-up and continuous improvement.	Saleable product and repeatable quality system.	Customer qualification and environmental/safety operating compliance are maintained.

Recommended early pilot scope

Do not begin with a fully integrated plant. Secure three to five representative quartz composites from distinct geological domains, build a bench/pilot beneficiation route, then conduct controlled reduction campaigns at an established silicon-metal facility. Send representative MG-Si samples to a specialist chlorosilane/polysilicon partner for impurity-removal trials. This sequence tests the critical feedstock thesis while limiting initial capital exposure.

Key go/no-go tests

- Does the high-purity quartz create measurably lower B/P or metal burden in MG-Si versus a lower-cost benchmark feed?
- Does that difference materially improve purification recovery, distillation duty, off-spec frequency or customer acceptance?
- Can the process hold repeatable quality across geological variability and multiple operating campaigns?
- Can the company secure enough qualified feedstock and reliable low-carbon power at a competitive delivered cost?
- Can the plant safely handle CO, hydrogen, HCl and chlorosilanes with acceptable environmental performance?

13. Commercial logic and principal risks

The business case should be built around the cost per kilogram of accepted electronic-grade polysilicon, not the cost per tonne of mined quartz or furnace silicon. Every additional process step should be evaluated by the value it creates at the final product boundary.

Where high-purity silica can create value

- Lower impurity loading into reduction or purification, if confirmed by mass-balance tests.
- More stable furnace burden and potentially improved operating consistency, if particle and mineral properties are well controlled.
- Reduced off-spec product or purification losses, if impurity behaviour is favourable.
- A second product stream in ultra-high-purity fused silica and quartz components, provided a separate quality and customer qualification programme is established.
- Traceable feedstock provenance and lot-level analytics that help demonstrate process control to customers.

Principal risks and mitigations

Risk	Potential consequence	Mitigation / evidence needed
HPQ premium has no downstream benefit	Higher feed cost without better EG-Si yield or quality.	Run matched pilot tests against lower-cost quartz; calculate cost per accepted kg.
B/P not adequately controlled	Electronic-grade specification failure.	Dedicated assays and impurity deportment study; qualify purification partner early.
Recontamination during handling	Loss of purity after beneficiation.	Qualified wear materials, clean transfer, packaging, blank controls and audit trail.
High energy and power-price exposure	Uncompetitive operating cost and carbon footprint.	Power strategy, energy recovery, efficiency guarantees and sensitivity analysis.
Complex chemical process hazards	Safety incidents, permitting delay or high compliance cost.	Technology licensor, inherently safer design, HAZOP/LOPA and full abatement.
Customer qualification takes longer	Delayed revenue and working-capital pressure.	Engage target crystal growers early; agree sampling and qualification protocol before plant design.
Technology route changes	Stranded capital or redesign.	Modular plant design; keep electrolysis and FBR as gated options.

Commercial recommendation

The first phase should be a development partnership model: SilicaFlux controls feedstock selection, beneficiation, analytics and project integration; specialist partners provide furnace campaigns, chlorosilane purification, polysilicon deposition and crystal qualification. Build proprietary capability only where data shows it creates a defensible advantage.

14. Recommended SilicaFlux configuration

The proposed company should be organised around one primary product and two adjacent options.

Business line	Product	Priority	Strategic role
SilicaFlux Silicon	Electronic-grade polysilicon	Core	Primary semiconductor materials business, initially using partner refining and CVD capacity.
SilicaFlux Quartz	High-purity silica and fused-quartz components	Adjacent	Potential earlier revenue path; independent product qualification and customer validation required.
SilicaFlux Process R&D;	Electrochemical silica reduction and process intensification	Long-term	Create differentiated low-carbon reduction technology after proof of purity, scale and cost.

First 12-month action plan

Workstream	First actions	Evidence expected
Feedstock	Secure representative samples from 3–5 candidate quartz bodies; assay trace elements and mineral inclusions.	Ranked feedstock database and preliminary mine-to-concentrate balance.
Beneficiation	Run bench tests on physical separation and selected leach/thermal options.	Flowsheet and concentrate specification supported by mass balance.
Reduction	Engage silicon-metal furnace partner; test at least two quartz/reductant combinations.	MG-Si yield, impurities, energy, gas and slag data.
Purification	Engage EG-Si producer / technology licensor to assess MG-Si samples and impurity profile.	Feasibility of TCS purification and CVD to customer-grade polysilicon.
Customers	Interview crystal growers and wafer producers; agree preliminary quality schedule and samples required.	Target specification, qualification route and indicative offtake requirements.
Engineering / EHS	Prepare concept site screening, energy options, chemical logistics, HAZID and environmental constraints.	Pre-FEED basis and major cost / permitting risks.

Final recommendation

SilicaFlux should not attempt to make semiconductor-grade silicon by stopping at the reduction furnace. The credible production chain is high-purity quartz beneficiation, metallurgical silicon production, chemical conversion and distillation, high-purity silicon deposition, and then crystal-growth qualification. The technical advantage must be demonstrated through impurity mass balance and cost per accepted kilogram. Start with a pilot and specialist partners; move to owned chemical purification and crystal assets only after product qualification and economics are proven.

Sources and technical notes

Sources were checked for this report on 2 November 2026. The references below support the process description, feedstock definitions, energy benchmarks and alternative technology assessment. Where a published value is used as an indicative benchmark, site-specific design and vendor guarantees take precedence.

[1] U.S. Geological Survey (USGS), “Silica Statistics and Information.” Defines high-purity quartz as less than 100 ppm total impurities (at least 99.99% SiO_2) and notes ultra-high-purity material at 10 ppm or less for demanding uses including fused-quartz crucibles for semiconductor manufacturing. USGS source.

[2] “A review of high-purity quartz for silicon production in Australia,” Australian Journal of Earth Sciences (2024), DOI: 10.1080/08120099.2024.2362296. Discusses lattice-bound impurities, the absence of one universal HPQ definition, and commonly cited element-specific screening benchmarks for quartz. DOI link.

[3] Shao et al., “Review of advanced purification technologies for high-purity quartz materials,” Materials Research Bulletin, Vol. 196 (2026), Article 113904, DOI: 10.1016/j.materresbull.2025.113904. Connects geological impurity profiles with HPQ processability and reviews purification methods. DOI link.

[4] Yu et al., “High-purity quartz sand from mineral purification: a review on process, mechanism, and environmental strategies,” Minerals Engineering, Vol. 235 (2026), Article 109801, DOI: 10.1016/j.mineng.2025.109801. Reviews physical purification, chemical treatment, heat treatment and environmental considerations. DOI link.

[5] Hoover, Bell & Rippy, “Emerging Technologies for Decarbonizing Silicon Production,” Journal of Sustainable Metallurgy 10, 1921–1932 (2024). Reports approximately 11–13 kWh/kg silicon for conventional production and reviews biocarbon, CCS/CCU, metallothermic, hydrogen and molten-salt approaches; notes several alternatives are demonstrated mainly at laboratory scale. Open-access article.

[6] “Silicon electrowinning by molten salts electrolysis,” Frontiers in Chemistry (2023), DOI: 10.3389/fchem.2023.1133990. Reviews metallurgical silicon production by carbothermic reduction and molten-salt electrochemical approaches, including purity and scale-up challenges. Open-access article.

[7] “Advances in production and optimization of electronic-grade polysilicon: A review of modified Siemens and silane methods,” Solar Energy Materials and Solar Cells 283 (May 2025), Article 113446, DOI: 10.1016/j.solmat.2025.113446. Reviews precursor synthesis, distillation and deposition optimisation. DOI link.

[8] “Silicon Electrochemistry in Molten Salts,” Chemical Reviews 120(3), 1690–1709 (2020), DOI: 10.1021/acs.chemrev.9b00428. Reviews silica electro-deoxidation, silicon electrowinning and remaining opportunities/challenges in high-purity silicon and oxygen-evolving anodes. DOI link.

[9] “Purification Technologies for High-Purity Quartz: From Mineralogy to Applications,” Separation & Purification Reviews (2025), DOI: 10.1080/15422119.2025.2494076. Reviews impurity types and combined magnetic separation, flotation and leaching approaches. DOI link.

Technical notes and limitations

- This report is a process-development concept. It does not specify proprietary Siemens reactor design, chlorosilane operating recipes, detailed furnace operating set-points or detailed chemical handling procedures.
- All numerical impurity benchmarks are preliminary screening references unless explicitly stated as a measured product specification.
- Theoretical mass-balance values assume pure silica and the simplified net reaction. Industrial consumption and yield must be established by pilot testing.
- Energy and emissions depend strongly on process configuration, electricity mix, boundary definition and recycle integration. Compare like-for-like boundaries.
- Electronic-grade material must be qualified by the intended crystal grower or semiconductor customer using their approved methods and limits.

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End of report